

10.11 Railway Emergency Communication

10.11.1 Introduction

- 10.11.1.1 Railway emergency communication (REC) is used for communication during an emergency situation, it is considered as a high priority communication and has distinct means for set-up and presentation. (I)
- 10.11.1.2 The implementation of REC is based on operational rules in the specific railway environment. (I)
- 10.11.1.3 The behavior for call handling is defined by operational rules and the operational rules are reflected in parameters and values in the common functions. (I)
- 10.11.1.4 The REC FRMCS application and the common functions are configurable in order to fulfil local operational rules. (I)
- 10.11.1.5 A REC consists of a REC-alert (which is always set up) including an optional information (e.g. text, voice prompts). After the REC-alert has been set up, one or more subsequent optional REC-voice or one or more subsequent optional REC-data communication(s) can be added, as shown in Figure 10-4. Please note that during the migration phase the REC-voice is automatically initiated together with a REC-alert. After the migration phase the not automatically linking is optional. (I)

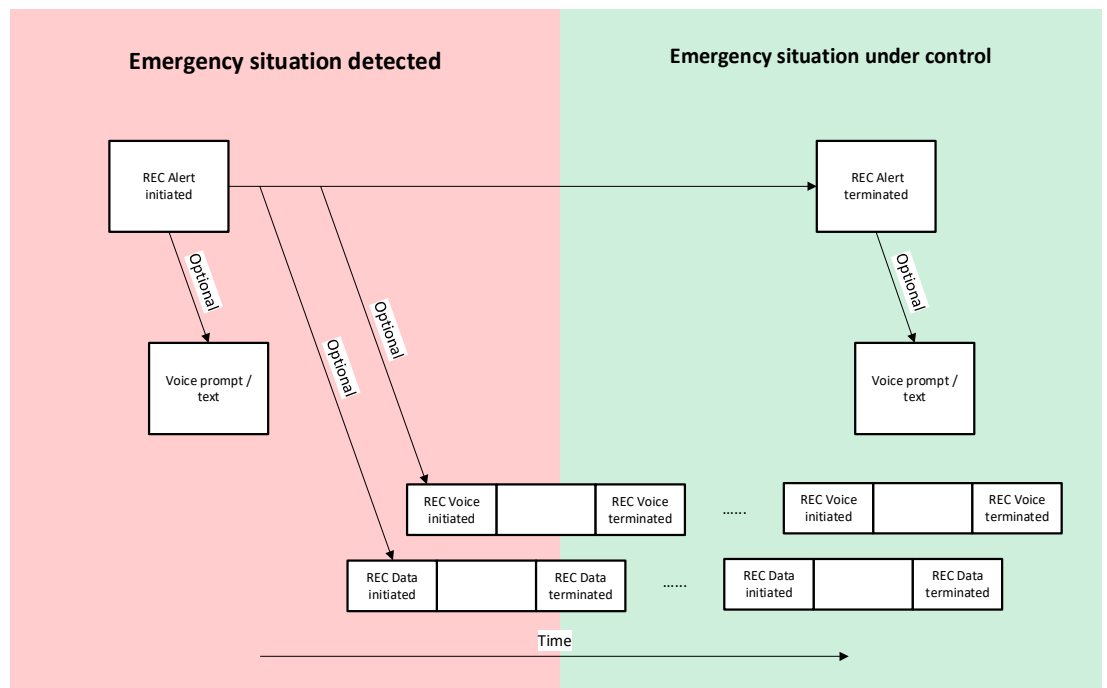


Figure 10-5: REC stages

- 10.11.1.5i In addition, mixing REC-voice and REC-data in the same REC-alert is useful for railway operation in emergency cases. (I)

- 10.11.1.6 The REC is based on the concept of predefined initiating area and predefined addressed area, as depicted in Figure 10-6. The initiating area corresponds to the track section where the REC is initiated by a user. This initiating area is linked to the danger location. Due to the reaction time of the moving user and due to other potential inaccuracies, the location of the danger will slightly differ from the place where the REC is initiated. In some cases, the danger location could even be located outside of the initiating area. Based on this initiating area, the FRMCS system will determine a corresponding predefined addressed area where the users are eligible to be alerted. The eligible users will be alerted depending on their functional identity and on their additional location information (train route setting, user's speed and user's direction). Associated to this addressed area, a list of predefined additional users (e.g. controllers) will be also alerted. (I).

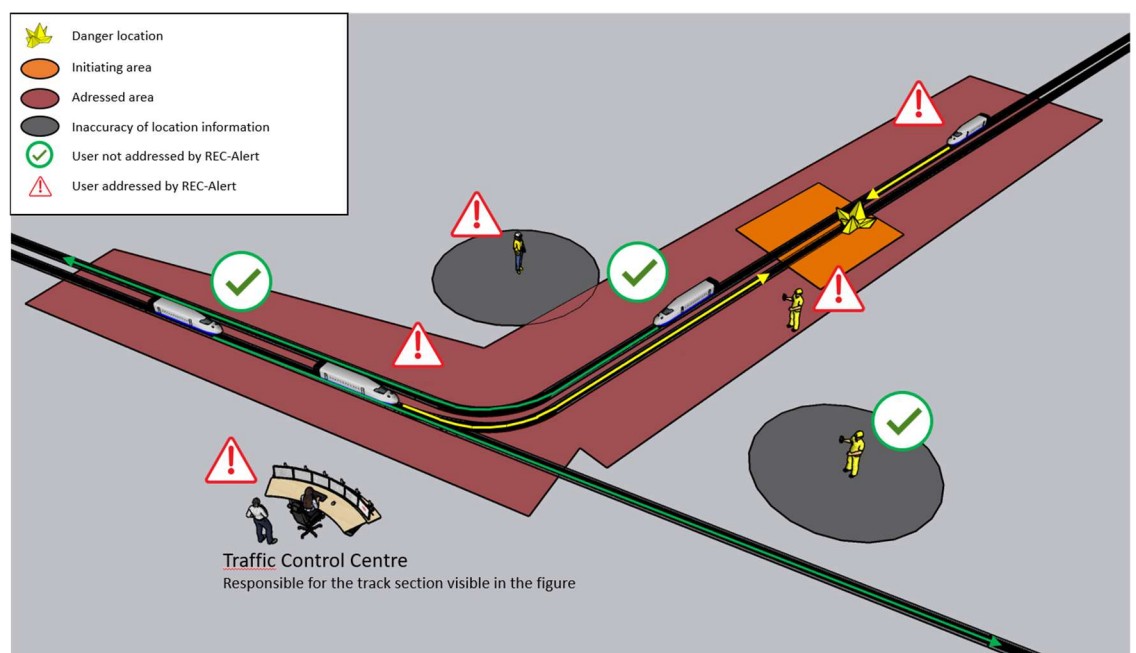


Figure 10-6: REC concept overview

- 10.11.1.7 The list of users addressed by the REC is determined and continuously adapted based on the predefined addressing rules for which a safety margin is applied. The addressing rules can be adjusted (fine-tuned) further on by the controller when knowledge of the actual emergency situation evolves. (I)
- 10.11.1.8 The size and shape of the initiating and addressed areas are defined by the infrastructure manager, based on various criteria like line speed, track characteristics, reaction time of the user as well as type of location information available for the user (position, direction, train route setting...) and their corresponding level of accuracy. (I)
- 10.11.1.9 Intentionally deleted.
- 10.11.1.10 For some users (e.g. train driver, trackside worker or shunter), the location information available for the initiating user's is used to determine the corresponding initiating area. (I)

- 10.11.1.11 For some users (e.g. controller, sensor-based system), the initiating area or addressed area is provided by the initiating's user. (I)
- 10.11.1.12 REC-alert is considered to be a data communication utilising the data communication function. (I)
- 10.11.1.13 Being involved in a REC-alert does not prevent a user from initiating a new REC-alert as long as allowed by the arbitration common function (I)
- 10.11.1.14 Being involved in a REC-alert does not prevent a user from initiating another communication, as long as allowed by the arbitration common function (I)
- 10.11.1.15 Intentionally deleted.
- 10.11.1.16 Intentionally deleted.
- 10.11.1.17 The optional REC-data communication can include, for example, pre-recorded messages, pictures, etc. (I)
- 10.11.1.18 An external system is able to initiate a REC. The initiating method to be used and the selected area(s) are similar to a driver (moving detection system, e.g. collision detector) or to a controller (stationary detection system, e.g. railway crossing obstruction detector). Additional information (e.g. text, voice prompts) can be included in the REC-alert. It is assumed that REC-voice or REC-data is not invoked by the external system. Usually the controller is terminating this REC. (I)
- 10.11.1.19 The Railway Emergency Communication application is comprehensively captured in this FRMCS application section. (I)
- 10.11.2 [Generic requirements](#)
- 10.11.2.1 A controller shall be able to initiate a REC-alert. (M)
- 10.11.2.2 A driver shall be able to initiate a REC-alert. (M)
- 10.11.2.3 Intentionally deleted.
- 10.11.2.4 A controller shall be able to initiate a REC-alert by selecting one addressed area. (M)
- 10.11.2.4a A controller shall be able to initiate a REC-alert by selecting several addressed areas. (M-Vx)
- 10.11.2.4i When a controller initiates a REC-alert by selecting several addressed areas, the resulting addressed area shall be the union of the selected addressed areas. (M-Vx)
- 10.11.2.5 A controller shall be able to initiate a REC-alert by selecting one initiating area. (M-Vx)
- 10.11.2.5i Intentionally deleted
- 10.11.2.5ii A controller shall be able to initiate a REC-alert by providing a specific train functional identity. (M-V3)

- 10.11.2.6 When a controller initiates a REC-alert by providing a specific train functional identity, the addressed area shall be based on the initiating area where this train is located. (M-V3)
- 10.11.2.6i A controller should be able to initiate a REC-alert by manually defining the addressed area (e.g. by drawing an addressed area on a map or by selecting track sections with start and end location). (O-V3)
- 10.11.2.6ii When a controller initiates a REC-alert by manually defining the addressed area, the addressed controllers shall be all the controllers associated to the predefined addressed areas fully or partially included in the manually defined area. (M-V3)
- 10.11.2.7 For a driver initiated REC-alert, the geographical position of the train shall be used to retrieve the initiating area. (M)
- 10.11.2.7ai For a driver initiated REC-alert, the user's geographical position shall be the front of the train. (M-Vx)
- 10.11.2.7i For a driver-initiated REC-alert, one or a combination of the following location information elements of the initiating driver's should be used to retrieve the initiating area: (O-Vx)
 - a) Railway infrastructure element(s) where the initiator is located (e.g. track section ID, station ID, signal box ID, track kilometer marking);
 - b) Other infrastructure element(s) where the initiator is located.
- 10.11.2.8 The REC-alert shall address the following users:
 - a) Drivers for which the train is located inside the addressed area; (M)
 - b) Intentionally deleted.;
 - c) Predefined controller(s) for the addressed area by its functional identity. (M)
 - d) Other users which are located inside the addressed area and for which the function label matches criteria set by the infrastructure manager. (M-V3)
- 10.11.2.8a If the driver has not yet registered to the train identity, the driver shall be addressed using a vehicle identity including driver function label. (M)
- 10.11.2.8ai The REC-alert shall address the driver for which the front of the train geographical position is located inside the addressed area. (M-Vx)
- 10.11.2.8i The addressing of the REC-alert should allow to exclude eligible users based on one or a combination of the following conditions: (O-Vx)
 - a) Train route setting;
 - b) Direction;
 - c) Speed.
- 10.11.2.8ii The excluded eligible users of the REC-alert could be for example: (I)
 - a) Train route setting: train is routed on a diverging train route from the initiating area;
 - b) Direction: train moving away from the initiating area;
 - c) Speed: low speed or stopped trains.

- 10.11.2.8iii After the initiation phase, the initiator of the REC-alert shall be considered also as an addressed user. (M)
- 10.11.2.9 It shall be possible to include at setup of the REC-alert additional information (e.g. text, voice prompts) that are permanently presented to the recipients. (M)
- 10.11.2.9i It shall be possible during an ongoing REC-alert to change the additional information (e.g. text, voice prompts) that are permanently presented to the recipients. (M-Vx)
- 10.11.2.9ii It shall be possible during an ongoing REC-alert, if no additional information was inserted at setup, to add the additional information (e.g. text, voice prompts) that are permanently presented to the recipients. (M-Vx)
- 10.11.2.10 It should be possible for a controller to change the addressed area(s) during an ongoing REC-alert. (O-Vx)
- 10.11.2.11 The FRMCS system shall monitor the users eligibility to participate in the REC-alert and adapt the recipients accordingly. (M)
- 10.11.2.12 Intentionally deleted.
- 10.11.2.13 Intentionally deleted.
- 10.11.2.14 The ongoing REC-alert shall be permanently indicated to the recipients. (M)
- 10.11.2.15 An addressed user no longer eligible for the ongoing REC-alert shall be informed and disconnected from the REC-alert. (M)
- 10.11.2.16 As long as the REC-alert is ongoing, the controller shall be informed about the relevant identity of the recipients currently addressed by a REC-alert. (M)
- 10.11.2.17 Intentionally deleted.
- 10.11.2.18 Intentionally deleted.
- 10.11.2.19 Intentionally deleted.
- 10.11.2.19i The controller shall be informed about the relevant identity of the users no longer addressed by the REC-alert. (M-Vx)
- 10.11.2.20 A driver already addressed by a REC-alert shall be able to initiate another REC-alert. (M)
- 10.11.2.21 Intentionally deleted.
- 10.11.2.22 A controller already addressed by REC-alert shall be able to initiate another REC-alert. (M)
- 10.11.2.22i It shall be possible for a controller to merge two ongoing REC-alerts. (M-Vx)
- 10.11.2.22ii After merging of two REC-alerts, the addressed users of the resulting REC-alert shall be the union of users that would have been addressed by each of the initial REC-alerts. (M-Vx)
- 10.11.2.22iii A driver shall not be allowed to leave a REC-alert. (M)
- 10.11.2.22iv In the case the initiating driver is moving outside the addressed area of the REC-alert, the ongoing REC-alert shall not be terminated. (M)

- 10.11.2.22v When the initiating driver is moving outside the addressed area of the REC-alert, the initiating driver shall be disconnected and no more be addressed by this REC-alert. (M)
- 10.11.2.22vi An authorised controller shall be allowed to leave the REC-alert. (M)
- 10.11.2.22vii A controller shall be able to re-join an ongoing REC-alert that the controller has previously left. (M-V3)
- 10.11.2.22viii An authorised controller shall be able to terminate the REC-alert. (M)
- 10.11.2.22ix The initiating driver shall be able to terminate the REC-alert, if allowed by configuration. (M-V3)
- 10.11.2.23 The FRMCS system shall support the presentation of additional information (e.g. text, voice prompts) to the addressed users upon termination of the REC-alert. (M-Vx)
- 10.11.2.23i The FRMCS system shall support an IM system administrator to predefine whether a REC-alert initiated by a driver is associated to a REC-voice or REC-data. (M-Vx)
- 10.11.2.23ii The FRMCS system shall support an IM system administrator to predefine whether a REC-alert initiated by a controller is associated to a REC-voice or REC-data. (M-Vx)
- 10.11.2.24 The FRMCS system shall support to predefine if a REC-voice shall be initiated for a given initiating function label, the outcome being the following:
 - a) REC-voice is set up automatically upon initiation of the REC-alert; (M-V3)
 - b) No REC-voice set up automatically upon initiation of the REC-alert. (M-Vx)
- 10.11.2.25 The FRMCS system shall support to predefine if a REC-voice shall be initiated for a given initiating area, the outcome being the following:
 - a) REC-voice is set up automatically upon initiation of the REC-alert; (M-V3)
 - b) No REC-voice set up automatically upon initiation of the REC-alert. (M-Vx)
- 10.11.2.26 In the case of using function label and initiating area to determine if a REC-voice has to be automatically initiated upon REC-alert initiation, the FRMCS system shall ensure that the configuration leads to a predictable outcome. (M-Vx)
- 10.11.2.26i A driver addressed by a REC-alert with no ongoing associated REC-voice shall be able to initiate a REC-voice associated to this REC-alert. (M-Vx)
- 10.11.2.26ii A controller addressed by a REC-alert with no ongoing associated REC-voice shall, when authorised, be able to initiate a REC-voice associated to this REC-alert. (M-Vx)
- 10.11.2.26iii Upon initiation of a REC-voice, the user(s) addressed by the REC-voice shall be those addressed by the associated REC-alert. (M)

- 10.11.2.26iv As soon as a user is addressed by a REC-alert with an ongoing REC-voice, this user shall also be addressed by the associated REC-voice. (M)
- 10.11.2.26v As long as a user is addressed by a REC-alert with an ongoing REC-voice, this user shall also be addressed by the associated REC-voice. (M)
- 10.11.2.26vi When a user is no longer eligible for receiving the REC-alert, and there is an ongoing associated REC-voice communication, the FRMCS system shall support to predefine whether:
 - a) REC-voice remains connected to this user; (M-Vx)
 - b) REC-voice is no more addressed to this user. (M)
- 10.11.2.26vii In the case of merging two REC-alerts, the associated active REC-voice communications shall also be merged. (M-Vx)
- 10.11.2.26viii A driver shall be able to leave a REC-voice, if allowed by configuration. (M-V3)
- 10.11.2.26ix An authorised controller shall be allowed to leave the REC-voice. (M)
- 10.11.2.26x The initiating driver of a REC-voice shall be able to terminate this REC-voice. (M-V3)
- 10.11.2.27 The FRMCS system shall support an IM system administrator to predefine for each function label and per initiating area, if a REC-data is set up automatically upon initiation of the REC-alert. (M-Vx)
- 10.11.2.28 During an ongoing REC-alert with no ongoing REC-data, a driver shall be able to initiate a REC-data associated to the REC-alert. (M-Vx)
- 10.11.2.28i During an ongoing REC-alert with no ongoing REC-data, an authorised controller shall be able to initiate a REC-data associated to the REC-alert. (M-Vx)
- 10.11.2.29 Intentionally deleted.
- 10.11.2.29i Upon initiation of the REC-data, the user(s) addressed by the REC-data shall be those addressed by the associated REC-alert. (M-Vx)
- 10.11.2.29ii As soon as a user is addressed by a REC-alert with an ongoing REC-data, this user shall also be addressed by the associated REC-data. (M-Vx)
- 10.11.2.29iii As long as a user is addressed by a REC-alert with an ongoing REC-data, this user shall also be addressed by the associated REC-data. (M-Vx)
- 10.11.2.30 When the user is no longer eligible for receiving the REC-alert, and there is an ongoing associated REC-data communication, the FRMCS system shall support to predefine whether. (M-Vx)
 - a) REC-data remains connected to this user;
 - b) REC-data is no more addressed to this user.
- 10.11.2.31 When a REC-voice is active and upon termination of the associated REC-alert, the FRMCS system shall predefine for a given initiating area the outcome being one of the following:
 - a) REC-voice communication remains active; (M-Vx)

- b) REC-voice communication is terminated. (M-V3)
- 10.11.2.31i When a REC-alert is active and upon termination of the associated REC-voice, the FRMCS system shall predefine for a given initiating area the outcome being one of the following:
 - a) REC-alert communication remains active; (M-Vx)
 - b) REC-alert communication is terminated. (M-V3)
- 10.11.2.31ii When a REC-voice is active and upon termination of the associated REC-alert, the FRMCS system shall predefine for a given function label the outcome being one of the following:
 - a) REC-alert communication remains active; (M-Vx)
 - b) REC-alert communication is terminated. (M-V3)
- 10.11.2.31iii When a REC-alert is active and upon termination of the associated REC-voice, the FRMCS system shall predefine for a given function label the outcome being one of the following:
 - a) REC-alert communication remains active; (M-Vx)
 - b) REC-alert communication is terminated. (M)
- 10.11.2.32 When a REC-data is active and upon termination of the associated REC-alert, the FRMCS system shall predefine the outcome being one of the following: (M-Vx)
 - a) REC-data communication remains active;
 - b) REC-data communication is terminated.
- 10.11.2.33 A REC-alert shall be recorded including its related metadata. (M-V3)
- 10.11.2.34 When a REC-voice is active, the following shall be recorded.
 - a) REC-voice communication content; (M-V3)
 - b) REC-voice communication related metadata. (M)
- 10.11.2.35 When a REC-data is active, the REC-data communication and its related metadata shall be recorded. (M-Vx)
- 10.11.2.36 The FRMCS system shall support an IM system administrator to configure georeferenced polygons to define the initiating areas used for REC-alert. (M-V3)
- 10.11.2.37 The FRMCS system should support an IM system administrator to configure georeferenced polyhedrons to define the initiating areas used for REC-alert. (O-Vx)
- 10.11.2.38 The use of polyhedrons could allow to discriminate between a track line and another one which overhangs the first one. (I)
- 10.11.2.38i The FRMCS system shall support an IM system administrator to configure georeferenced polygons to define for each initiating area a corresponding addressed area user for REC-alert. (M-V3)
- 10.11.2.39 The FRMCS system should support an IM system administrator to configure georeferenced polyhedrons to define for each initiating area a corresponding addressed area used for REC-alert. (O-Vx)